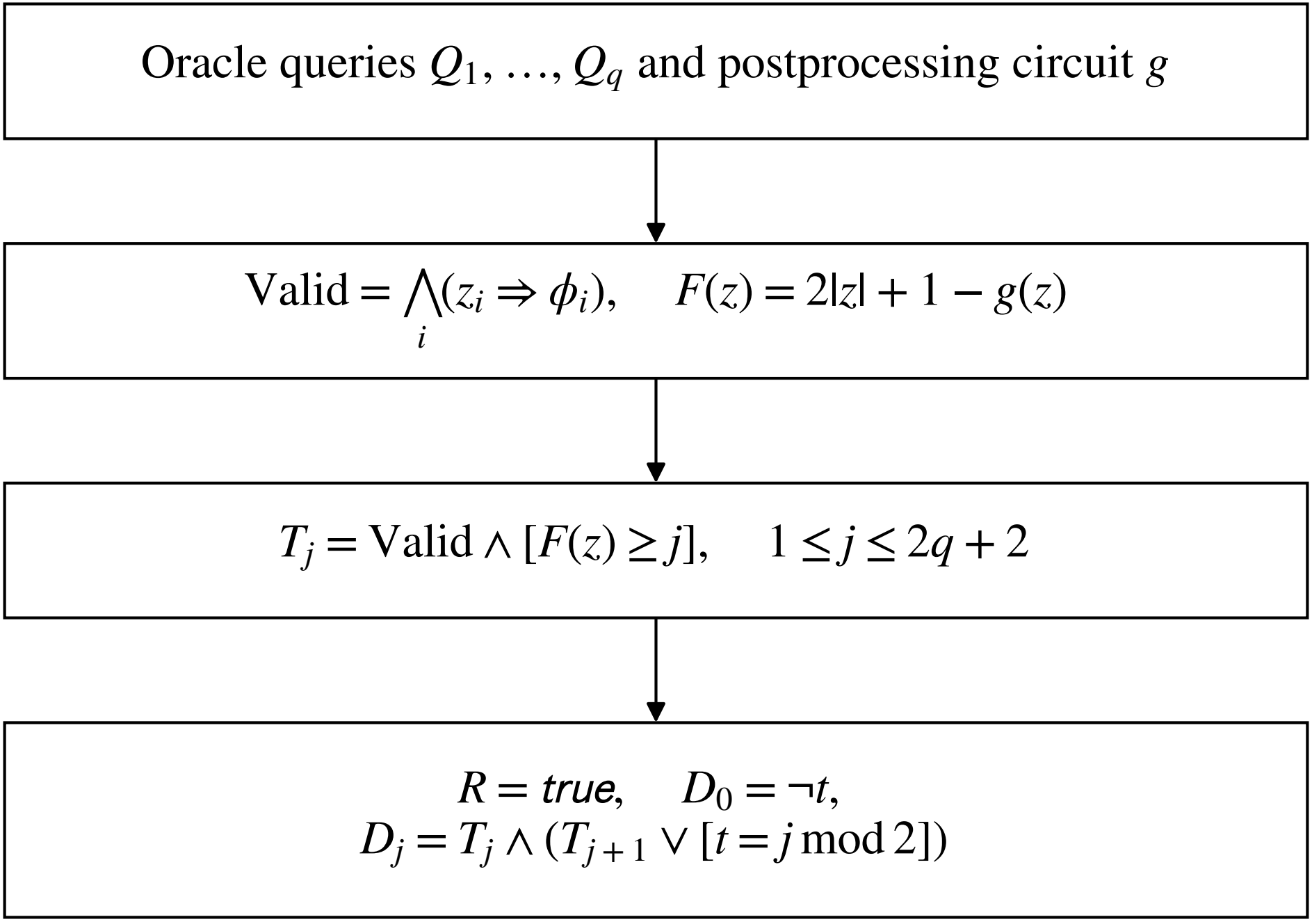
Oracle queries $Q_1, \ldots, Q_q$ and postprocessing circuit g
$\text{Valid} = \bigwedge_i (z_i \Rightarrow \phi_i), \quad F(z) = 2|z| + 1 - g(z)$
$T_j = \text{Valid} \wedge [F(z) \geq j], \quad 1 \leq j \leq 2q + 2$
$R = \mathit{true}, \quad D_0 = \neg t,$
$D_j = T_j \wedge (T_{j+1} \vee [t = j \bmod 2])$